Machine Learning

DSI61133

Overview of Machine Learning

Learning Objectives

01 Students are able to explain the fundamental concepts and goals of machine learning

02 Students are able to identify the main types of machine learning

AGENDA

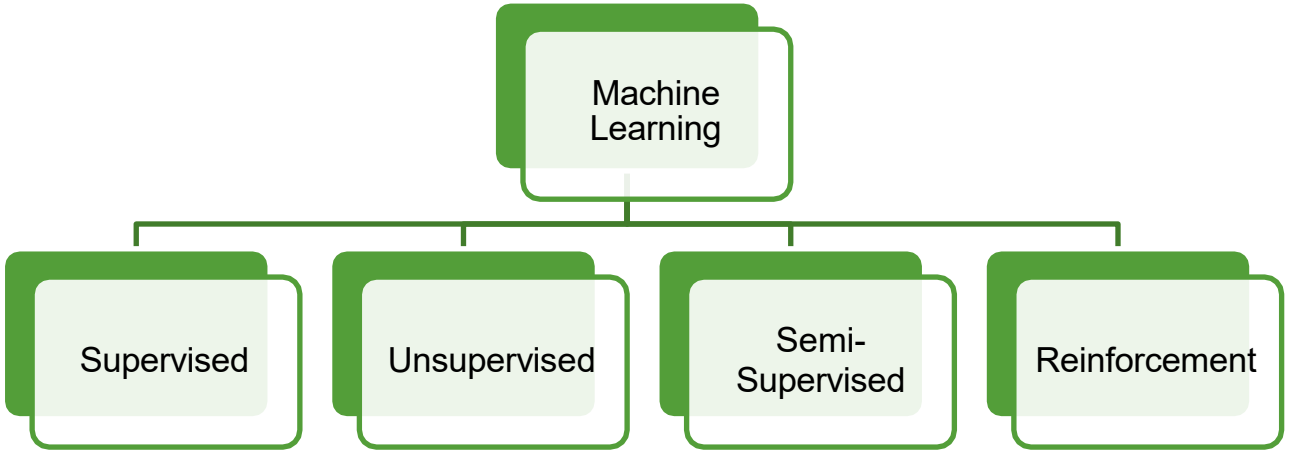
- Basic Concepts of Machine Learning
- Basic Concepts of Supervised Learning
- Basic Concepts of Unsupervised Learning
- Basic Concepts of Reinforcement Learning
- Basic Concepts of Ensemble Learning

“Machine learning is the science of getting computers to act without being explicitly programmed.”

- Arthur Samuel

Overview of Machine Learning

Overview of Machine Learning



Overview of Machine Learning

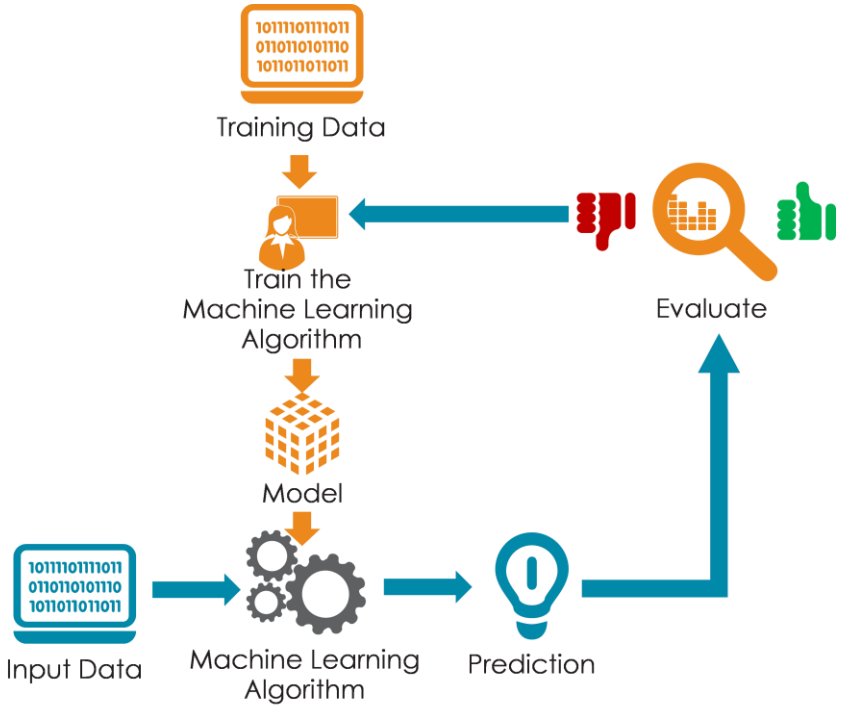
Pre - Processing

Overview of Machine Learning

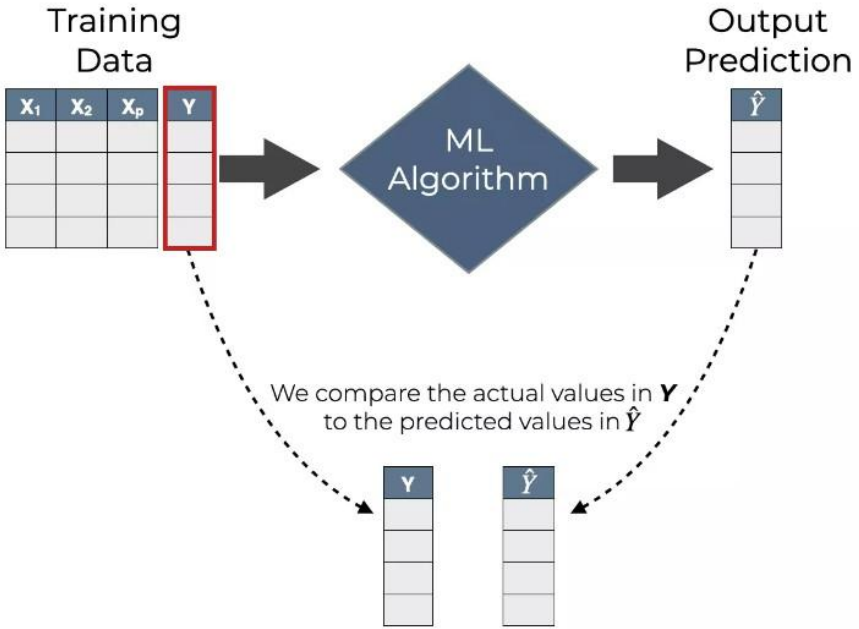
Name	Score	Presence	Status
Andi	85	90%	Passed
Budi	70	0.8	L
Citra		85	1
Dedi	60	75%	0
Eka	95	110%	Passed
Andi	85	90%	Passed

Imagine you're a data analyst. Before building a model, try to identify all the issues in the data.

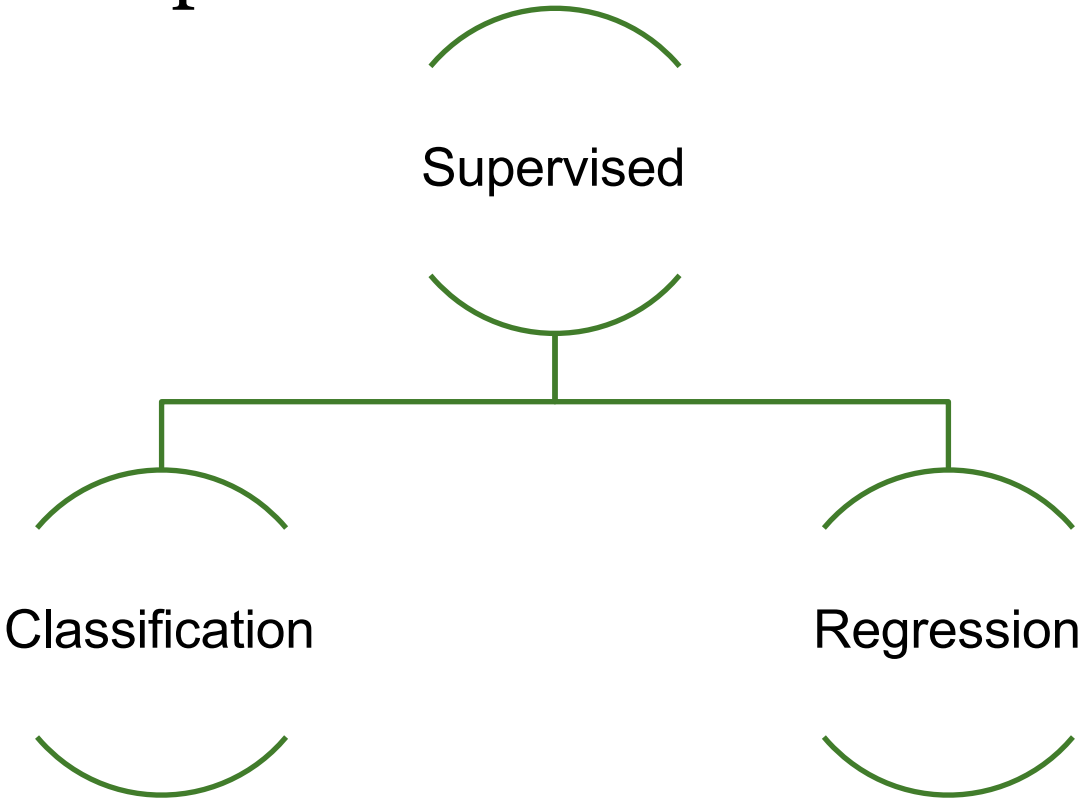
Overview of Machine Learning



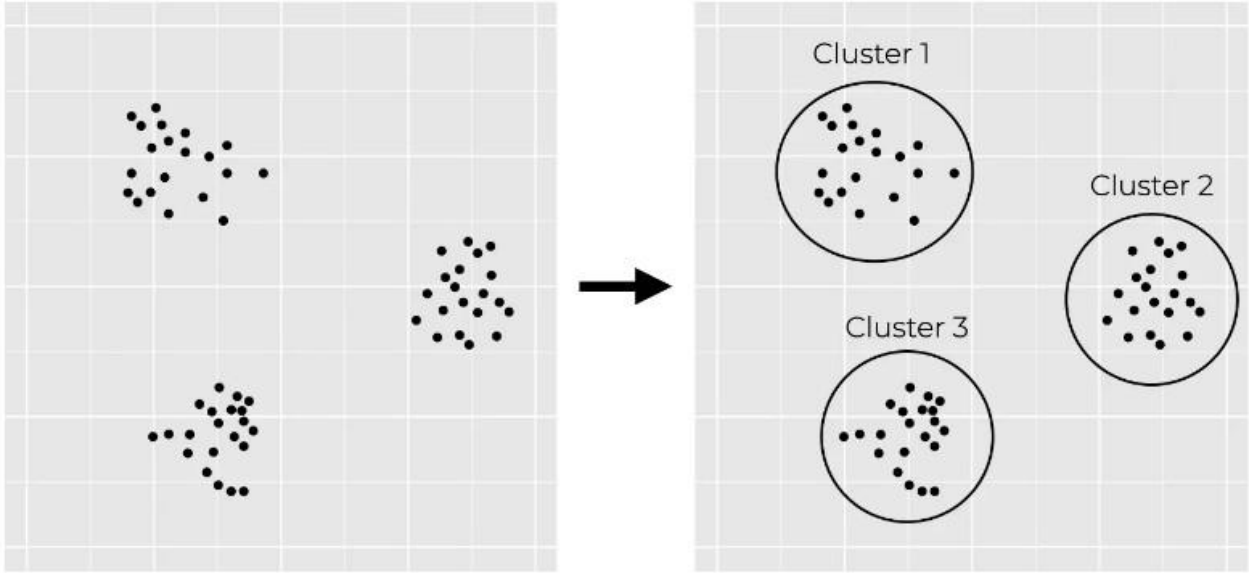
Supervised Machine Learning



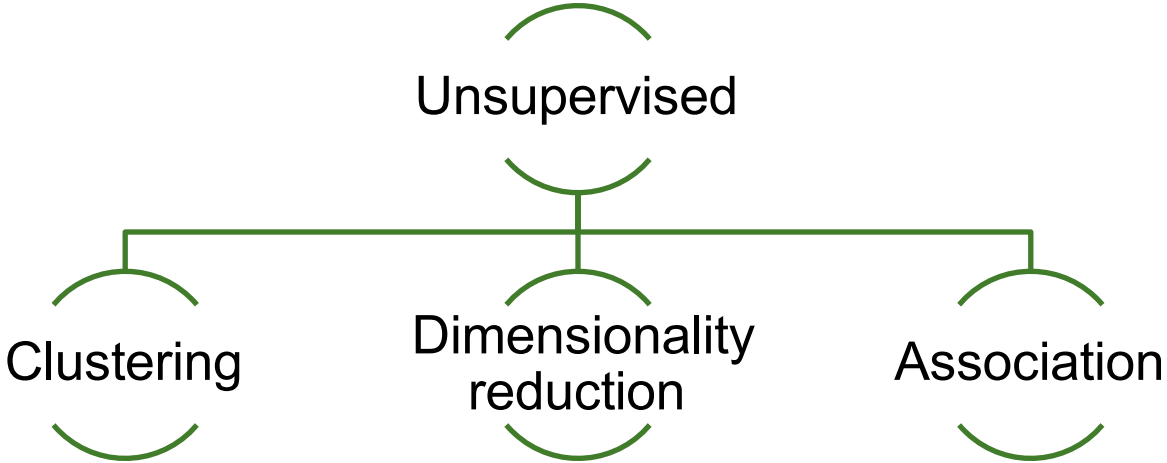
Supervised Machine Learning



Unsupervised Machine Learning



Unsupervised Machine Learning



How to resolve this?

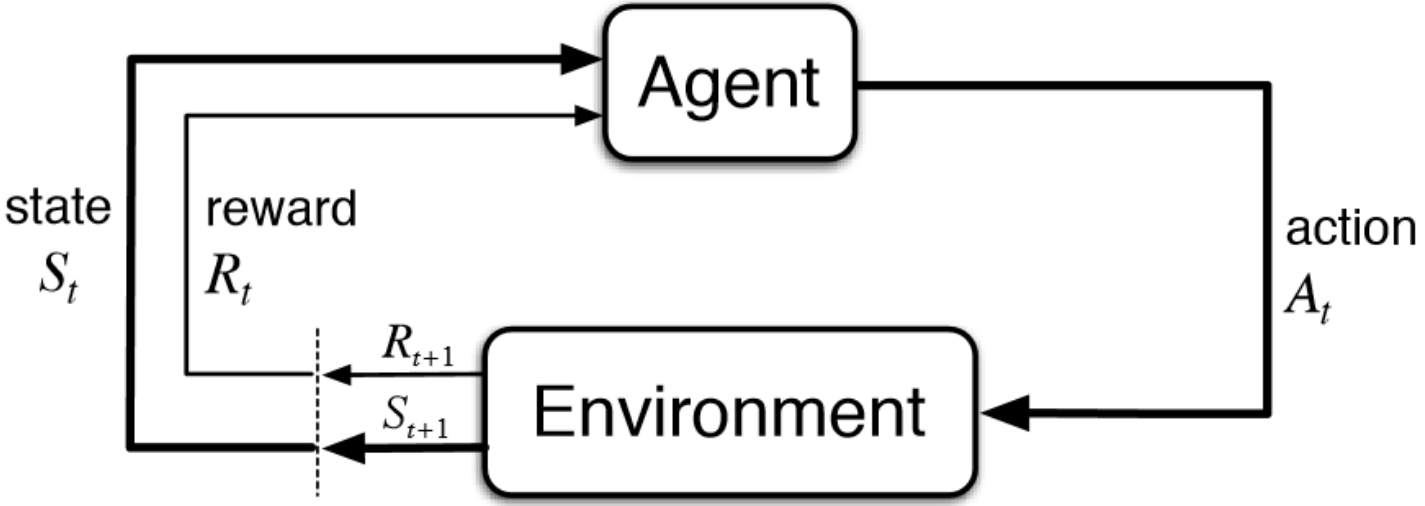
X_1	X_2	X_3	X_p	Y
				✓
				missing
				✓
				✓

Here, some of the values in the target variable Y are missing

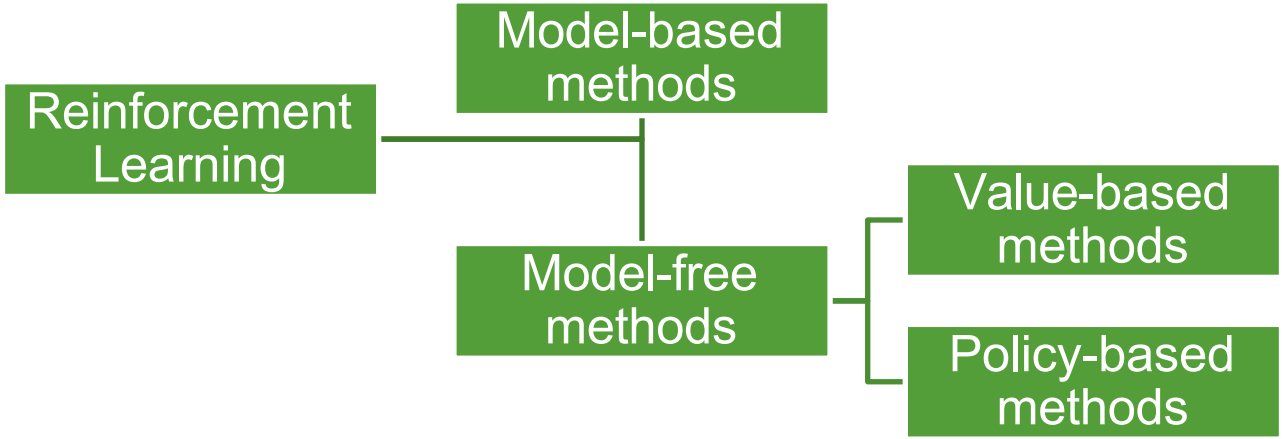
Basic Concepts of Reinforcement Learning

How does **Reinforcement Learning** work?

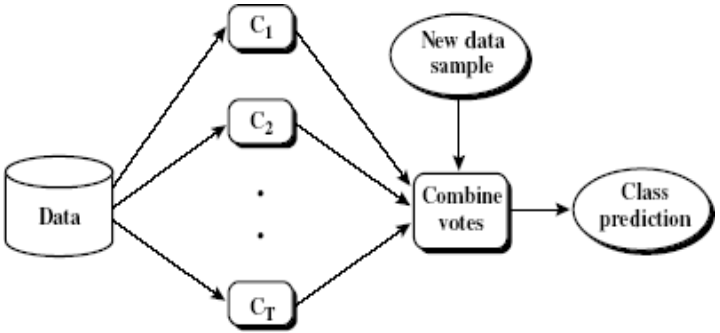
Basic Concepts of Reinforcement Learning



Reinforcement Learning Methods



Ensemble Methods: Increasing the Accuracy



- Ensemble methods
  - Use a combination of models to increase accuracy
  - Combine a series of k learned models, M_1, M_2, …, M_k, with the aim of creating an improved model M*
- Popular ensemble methods
  - Bagging: Parallel models + averaging/voting
  - Boosting: Sequential learning to correct errors
  - Stacking: Combining multiple model outputs

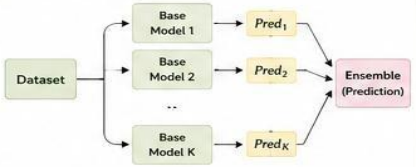
VISUALIZATION OF ENSEMBLE METHODS

Combining multiple models to achieve better performance, robustness, and generalization.

Goal: Visualize how ensemble methods combine multiple models to improve predictive performance.

1 WHAT IS ENSEMBLE METHOD?

Ensemble methods combine predictions from multiple base models to obtain a stronger model with better generalization.



Many models, one goal: better predictions!

2 WHY USE ENSEMBLE?

Higher Accuracy
Reduces bias and variance.

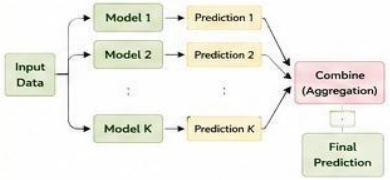
More Robust
Less sensitive to noise and outliers.

Better Generalization
Performs well on unseen data.

Stability
More consistent and reliable predictions.

3 HOW ENSEMBLE WORKS (IN GENERAL)

Each base model makes a prediction.
The ensemble combines them to produce the final prediction.



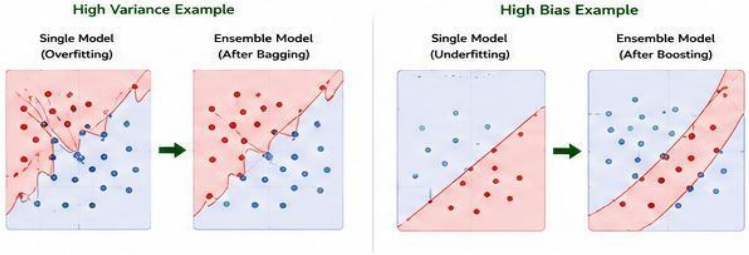
Common aggregation: Average / Majority Vote / Weighted Average

4 MAIN TYPES OF ENSEMBLE METHODS

Type	Description	Example Algorithms	Key Idea	Variance	Bias
Bagging (Parallel)	Train models independently on different bootstrap samples and aggregate.	Random Forest, Bagging, Extra Trees	Reduce variance	↓	≈
Boosting (Sequential)	Train models sequentially, each focusing on the errors of the previous.	AdaBoost, Gradient Boosting, XGBoost, LightGBM	Reduce bias and variance	↓	↓
Stacking (Meta-Learning)	Train a meta-model to learn how to best combine the base models.	Stacked Generalization, Blending	Capture different strengths	↓	↓
Voting (Simple Ensemble)	Combine predictions by majority vote (classification) or average (regression).	Hard Voting, Soft Voting	Simple and effective	↓	≈

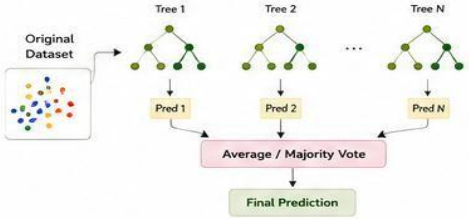
↓ = decreased, ↑ = increased, ≈ = no significant change

5 VISUAL COMPARISON: SINGLE MODEL vs ENSEMBLE



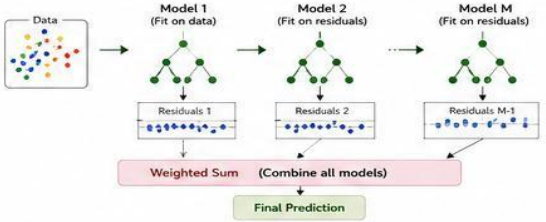
6 EXAMPLE: RANDOM FOREST (BAGGING)

Many decision trees are trained on different bootstrap samples and random feature subsets.

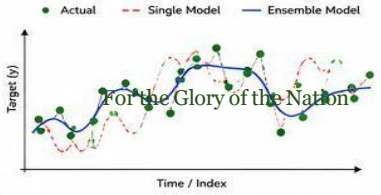


7 EXAMPLE: GRADIENT BOOSTING

Models are trained sequentially. Each new model focuses on the errors of the previous ensemble.



8 ENSEMBLE IN ACTION: REGRESSION EXAMPLE



The ensemble model (blue) follows the true pattern better than a single model (red dashed).

Better fit, better results!

For the Glory of the Nation

9 WHEN TO USE ENSEMBLE?

- When single models have high variance (e.g., decision trees).
- When data is noisy or complex.
- When better generalization is needed.
- When you want more stable and reliable predictions.

Ensemble methods often provide the best "bang for your buck" in machine learning.

10 ADVANTAGES & LIMITATIONS

Advantages

- Improved accuracy and generalization
- More robust to overfitting
- Handles complex relationships
- More stable predictions

Limitations

- Higher computation cost
- Harder to interpret (less explainable)
- Requires more memory/storage
- May overfit if not tuned properly (e.g., boosting)

11 BEST PRACTICES

- Use cross-validation to assess ensemble performance.
- Tune hyperparameters for base models and ensemble.
- Choose the right ensemble type for your problem.
- Combine diverse models (different algorithms or features).
- Monitor for overfitting, especially in boosting.

The power of ensemble lies in diversity + aggregation. Diverse models make better ensembles.

ENSEMBLE METHODS IN MACHINE LEARNING

Many models, working together, are often better than one.

GOAL: *Leverage the power of multiple models to improve predictive performance, reduce variance, and increase generalization.*

1 BIAS–VARIANCE TRADE-OFF

Why Ensemble Methods?

High Bias (Underfitting)
Model too simple to capture the pattern.
Example: Linear model for complex data

High Variance (Overfitting)
Model too complex, sensitive to small changes in data.
Example: Deep decision

How Ensemble Helps

Bagging	Boosting	Stacking
Reduces Variance	Reduces Bias	Combines Strengths (Clever Combination)

Ensembles try to compensate for the weaknesses of single models.

2 WHY DECISION TREES ARE OFTEN USED?

Decision Trees

- Easy to understand
- Non-linear
- Do not require scaling
- ... but:
  - High variance
  - Easy to overfit
  - Sensitive to small changes in data

Perfect for Ensembles

Bagging (Random Forest)
Reduces variance by training many trees on **different data samples.**

Boosting (XGBoost, GBM)
Reduces bias by sequentially fixing errors.

A single tree may be weak, but a forest of trees is strong.

3 HARD VOTING vs SOFT VOTING

Hard Voting (Majority Class)
Choose the class with the most votes.

Model	Prediction
Model A	Spam
Model B	Spam
Model C	Not Spam

Result: Spam

Soft Voting (Average Probabilities)
Use predicted probabilities (more informative).

Model	Prob Spam
Model A	0.90
Model B	0.80
Model C	0.40

Average = 0.70
Result: Spam

Soft voting is usually better because it considers the confidence of each model.

4 WEIGHTED ENSEMBLE

Not all models are equally good.

Model	Accuracy (Example)	Weight	Influence
Random Forest	92%	0.50	High
XGBoost	90%	0.30	Medium
SVM	85%	0.20	Low

Weighted Average Formula

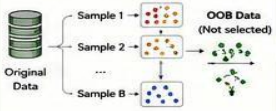
$$\hat{y} = \sum_{i=1}^{n} w_i \cdot \hat{y}_i$$

$\hat{y}$ = final prediction
$\hat{y}_i$ = prediction from model i
w_i = weight for model i

Better models get more influence in the final decision.

5 OOB (OUT-OF-BAG) ERROR IN RANDOM FOREST

Bootstrap Sampling



- ✓ Each tree is trained on a bootstrap sample.
- ✓ Data not selected = OOB data.

Benefits of OOB Error

- ✓ Used for internal model evaluation (no extra validation set needed)
- ✓ Provides an unbiased estimate of generalization error

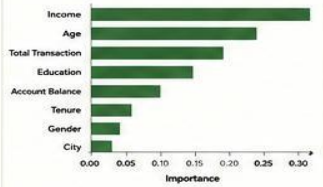
OOB Error Formula

$$OOB\ Error = \frac{1}{n}\sum_{i=1}^{n} L(y_i, \hat{y}_{OOB,i})$$

L = loss function (e.g., 0–1 loss, MSE, log loss)

6 FEATURE IMPORTANCE IN ENSEMBLE

Example: Random Forest Feature Importance



Helps us understand which features have the most impact on the prediction.

Benefits:

- ✓ Interpretability
- ✓ Feature selection
- ✓ Business insight

7 ENSEMBLE IS NOT ALWAYS BETTER

Limitations

- **More Complex** — Harder to understand and explain.
- **Slower Training** — Requires more time.
- **High Computational Cost** — Needs more CPU, RAM, and storage.
- **Lower Interpretability** — Difficult to explain to users or regulators.

Trade-off: Is the small improvement in accuracy worth the extra complexity and cost?

8 ENSEMBLE vs DEEP LEARNING

Tabular / Structured Data (Usually)
Ensemble methods (RF, XGBoost, etc.) often perform better.

Unstructured Data (Often)
Deep learning usually performs better.

Image · Audio · NLP

Choose the right tool based on data type and the problem.

9 CROSS VALIDATION FOR ENSEMBLE

Use robust evaluation for complex models.

K-Fold Cross Validation (k = 5)

	1	2	3	4	5	Result
Fold 1	Test	Train	Train	Train	Train	Score
Fold 2	Train	Test	Train	Train	Train	Score
Fold 3	Train	Train	Test	Train	Train	Score
Fold 4	Train	Train	Train	Test	Train	Score
Fold 5	Train	Train	Train	Train	Test	Score

Final Score
Average of 5 scores

Don't trust a single train-test split!

10 REAL-WORLD USE CASES

Domain	Problem	Ensemble Methods
Fraud Detection	Detecting fraudulent transactions	Random Forest, XGBoost
Credit Scoring	Predicting credit risk	Gradient Boosting
Recommender System	Predicting user preferences	Stacking, Blending
Medical Diagnosis	Disease classification	Boosting, Stacking
Kaggle Competitions	Winning data science competitions	Stacking Ensemble

Ensembles are widely used in industry and competitions because they deliver strong performance.

11 XGBOOST: WHY SO POPULAR?

- ✓ Optimized Boosting (fast & efficient)
- ✓ **Regularization** (L1 & L2) to prevent overfitting
- ✓ **Handles Missing Values**
- ✓ **Parallel Processing**
- ✓ **High Performance and Stable**

When to Use?
Suitable for many structured/tabular problems, especially when you need high accuracy.

XGBoost is widely used and often wins many data science competitions.

12 OVERFITTING IN ENSEMBLE?

Yes, ensemble can overfit!

How to Reduce It:

- ✓ Regularization
- ✓ Early Stopping
- ✓ Cross Validation
- ✓ Limit Model Complexity
- ✓ Use Enough Data



13 ENSEMBLE WORKFLOW (PRACTICAL)

Data → Preprocessing → Train Base Models → Combine Predictions → Evaluate → Tune Ensemble → Deploy

Iterate & improve until performance is optimal.

For the Glory of the Nation

14 KEY TAKEAWAYS

Choose the Right Method

- Bagging → Reduce variance
- Boosting → Reduce bias
- Stacking → Best combination

Keys to Success

- ✓ Model diversity
- ✓ Good quality data
- ✓ Proper evaluation
- ✓ Careful tuning

Remember!
Ensemble is not just about adding more models, but about combining models that complement each other.

> In modern machine learning, it's not about one model winning, but about the combination of several models that completes each other. ❞

Examples of Research Implementations

Real-world applications of Machine Learning from recent studies (2020–2025)

1 Hate Speech Detection in Indonesian Social Media

Wenando, F. A., et al. (2025). Optimizing Hate Speech Detection in Indonesian Social Media: An ADASYN and LSTM-Based Approach. *Journal Européen des Systèmes Automatisés*.

Real-World Problem

The increasing spread of hate speech on Indonesian social media platforms negatively impacts online discourse and social harmony.

Method / Approach

- ADASYN for handling class imbalance
- LSTM for text classification



Implementation Example

- Web or mobile application to detect hate speech in real time
- Moderation dashboard for social media content

2 Hate Speech Detection using FastText and LSTM

Wenando, F. A., N. Yusoff, and N. Irzin. (2024). Optimizing hate speech detection in the Indonesian language using FastText and LSTM algorithms. *2nd International Symposium on Information Technology and Digital Innovation (ISITDI)*. IEEE.

Real-World Problem

The lack of effective methods to detect hate speech in the Indonesian language, which has diverse and informal expressions.

Method / Approach

- FastText for word representation
- LSTM for sequence classification



Implementation Example

- Content monitoring system for social media platforms
- API for automatic hate speech detection

3 Generation of Synthetic Continuous Numerical Data

Aziira, A. H., N. A. Setiawan, and I. Soesanti: (2020). Generation of synthetic continuous numerical data using generative adversarial networks. *Journal of Physics: Conference Series*.

Real-World Problem

Limited availability of real numerical data due to privacy, cost, or access restrictions, which hinders data-driven research and application development.

Method / Approach

- Generative Adversarial Networks (GAN) for generating synthetic continuous numerical data



Implementation Example

- Data augmentation for limited datasets
- Privacy-preserving data sharing
- Synthetic data for education, research, or system testing

4 Classification of Indonesian Cyber Crime Cases

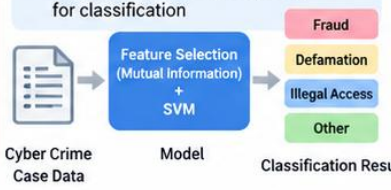
Rahmat, R. F., Aziira, A. H., et al. (2023). Classifying Indonesian Cyber Crime Cases under IT Law Using a Hybrid of Mutual Information and Support Vector Machine. *International Journal of Safety & Security Engineering*.

Real-World Problem

The need for accurate classification of Indonesian cyber crime cases to support law enforcement efforts under the Information and Electronic Transactions (IET) Law.

Method / Approach

- Mutual Information for feature selection
- Support Vector Machine (SVM) for classification



Implementation Example

- Decision support system for cyber crime classification
- Assistance for law enforcement in handling and categorizing cases

Thank You

For the Glory of the Nation